C **Foundations of Programming** Lab Assignments 16 – 22

A#16 – Even/Odd | A#17 – Sine Series | A#18 – Strings | A#19 – Employee | A#20 – Point

ASSIGNMENT NO. 19

Employee Structure in C

struct EMPLOYEE with name, designation, gender, DOJ, salary — compute totals, gender split, salary filter, designation filter.

C CODE

```c
#include <stdio.h>
#include <string.h>

struct EMPLOYEE {
    char  name[50];
    char  designation[50];
    char  gender[10];
    char  doj[15];
    float salary;
};

int main() {
    int n, i, male = 0, female = 0;

    printf("Enter number of employees: ");
    scanf("%d", &n);

    struct EMPLOYEE emp[n];

    for (i = 0; i < n; i++) {
        printf("\nEmployee %d:\n", i + 1);
        printf("  Name        : "); scanf("%s", emp[i].name);
        printf("  Designation : "); scanf("%s", emp[i].designation);
        printf("  Gender      : "); scanf("%s", emp[i].gender);
        printf("  Date of Joining: "); scanf("%s", emp[i].doj);
        printf("  Salary      : "); scanf("%f", &emp[i].salary);
        if (strcmp(emp[i].gender, "Male") == 0) male++;
        else female++;
    }

    printf("\nTotal employees   = %d\n", n);
```

```
    printf("Male employees    = %d\n", male);
    printf("Female employees  = %d\n", female);

    printf("\nEmployees with salary > 10000:\n");
    for (i = 0; i < n; i++)
        if (emp[i].salary > 10000)
            printf("  %s (%.2f)\n", emp[i].name, emp[i].salary);

    printf("\nEmployees with designation 'AsstManager':\n");
    for (i = 0; i < n; i++)
        if (strcmp(emp[i].designation, "AsstManager") == 0)
            printf("  %s\n", emp[i].name);

    return 0;
}
```

SAMPLE I/O

```
Emp 1: Ravi AsstManager Male 15000
Emp 2: Anita Clerk Female 8000

Total employees = 2
Male = 1   Female = 1
Salary > 10000: Ravi (15000.00)
AsstManager: Ravi
```

STRUCTURE FIELDS

NAME
char[50]

DESIGNATION
char[50]

GENDER
char[10] — Male/Female

DOJ
char[15] — DD-MM-YYYY

SALARY
float

FLOWCHART

Syntax error in text
mermaid version 10.9.1